

Tree-Level Cross Section for $Z'Z'\chi \rightarrow Z'\chi$

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1 Setup and Conventions

We consider a dark sector with a $U(1)_D$ gauge symmetry under which a Dirac fermion χ (dark matter) carries unit charge and Z' is the associated massive gauge boson. The relevant interaction Lagrangian is

$$\mathcal{L}_{\text{int}} = -g_D Z'_\mu \bar{\chi} \gamma^\mu \chi, \quad (1)$$

where $g_D = \sqrt{4\pi\alpha_X}$ is the dark gauge coupling.

Kinematics. We label the external momenta as

$$Z'(k_1) Z'(k_2) \chi(k_3) \rightarrow Z'(p_1) \chi(p_2), \quad p_2 = k_1 + k_2 + k_3 - p_1. \quad (2)$$

All particles are on-shell: $k_1^2 = k_2^2 = p_1^2 = m_{Z'}^2$, and $k_3^2 = p_2^2 = m_\chi^2$.

Invariants. With 5 external legs there are $5(5-3)/2 = 5$ independent Lorentz invariants. We define:

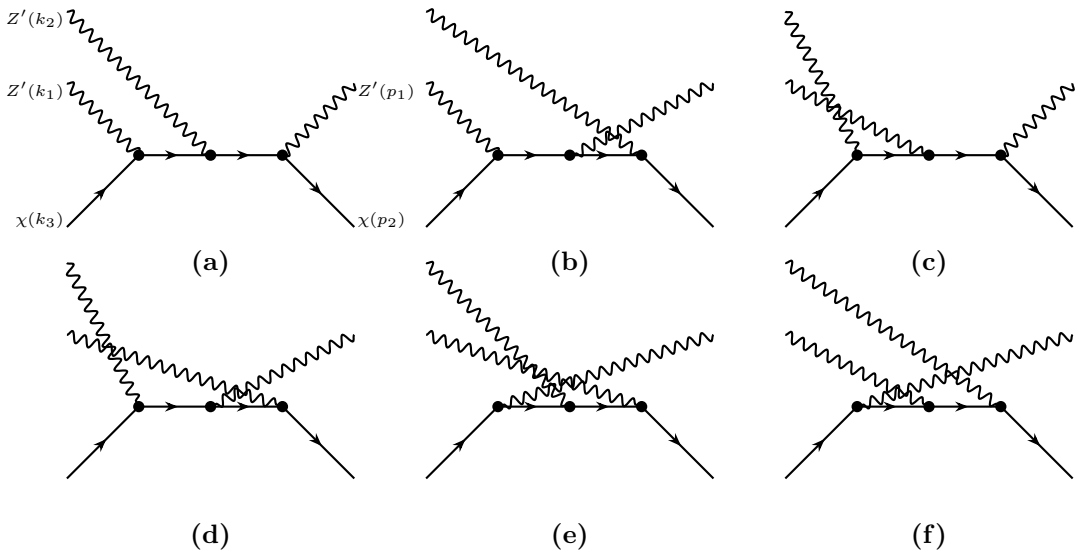
$$s_{12} \equiv (k_1 + k_2)^2, \quad s_{13} \equiv (k_1 + k_3)^2, \quad (3)$$

$$t_1 \equiv (k_1 - p_1)^2, \quad t_2 \equiv (k_2 - p_1)^2, \quad t_3 \equiv (k_3 - p_1)^2. \quad (4)$$

The sixth scalar product, $k_2 \cdot k_3$, is fixed by the on-shell constraint $p_2^2 = m_\chi^2$.

2 Feynman Diagrams

Since Z' couples only to the $\bar{\chi}\text{--}\chi$ current, every diagram has a single fermion line running from $\chi(k_3)$ to $\chi(p_2)$, with the three vector bosons $\{Z'(k_1), Z'(k_2), Z'(p_1)\}$ attached at three vertices along this line. The three vertex positions can be populated by any of the three bosons, giving $3! = 6$ permutations. The six diagrams are shown below with external legs at fixed positions. We label the vertices along the fermion line as v_1, v_2, v_3 from left to right. In each diagram the external legs are: $Z'(k_2)$ at upper far-left, $Z'(k_1)$ at upper near-left, $\chi(k_3)$ at lower-left, $Z'(p_1)$ at upper-right, and $\chi(p_2)$ at lower-right.



3 Amplitudes

Each vertex contributes a factor $(-ig_D\gamma^{\mu_i})$, where μ_i is the Lorentz index of the Z' boson attached at that vertex. Each internal fermion segment between vertices v_i and v_{i+1} contributes a propagator $i(\not{P} + m_\chi)/(P^2 - m_\chi^2)$, where P is the momentum flowing through that segment (determined by momentum conservation at each vertex). We denote by μ, ν, ρ the Lorentz indices carried by $Z'(k_1), Z'(k_2), Z'(p_1)$ respectively, regardless of which vertex they attach to.

Factoring out $(-ig_D)^3 = -ig_D^3$, the total amplitude is

$$\mathcal{M}^{\mu\nu\rho} = g_D^3 \sum_{i=a}^f \mathcal{A}_i^{\mu\nu\rho}, \quad (5)$$

where the six reduced amplitudes $\mathcal{A}_{(a)-(f)}^{\mu\nu\rho}$ read as follows.

Diagram (a): $Z'(k_1) \rightarrow v_1, Z'(k_2) \rightarrow v_2, v_3 \rightarrow Z'(p_1)$.

Internal momenta: $P_1 = k_3 + k_1, P_2 = k_3 + k_1 + k_2$.

$$\mathcal{A}_a^{\mu\nu\rho} = \frac{\bar{u}(p_2) \gamma^\rho (\not{P}_2 + m_\chi) \gamma^\nu (\not{P}_1 + m_\chi) \gamma^\mu u(k_3)}{D_1^{(a)} D_2^{(a)}} \quad (6)$$

$$D_1^{(a)} = (k_3 + k_1)^2 - m_\chi^2 = 2k_1 \cdot k_3 + m_{Z'}^2, \quad D_2^{(a)} = (k_3 + k_1 + k_2)^2 - m_\chi^2.$$

Diagram (b): $Z'(k_1) \rightarrow v_1, v_2 \rightarrow Z'(p_1), Z'(k_2) \rightarrow v_3$.

Internal momenta: $P_1 = k_3 + k_1, P_2 = k_3 + k_1 - p_1$.

$$\mathcal{A}_b^{\mu\nu\rho} = \frac{\bar{u}(p_2) \gamma^\nu (\not{P}_2 + m_\chi) \gamma^\rho (\not{P}_1 + m_\chi) \gamma^\mu u(k_3)}{D_1^{(b)} D_2^{(b)}} \quad (7)$$

$$D_1^{(b)} = (k_3 + k_1)^2 - m_\chi^2 = 2k_1 \cdot k_3 + m_{Z'}^2, \quad D_2^{(b)} = (k_3 + k_1 - p_1)^2 - m_\chi^2.$$

Diagram (c): $Z'(k_2) \rightarrow v_1, Z'(k_1) \rightarrow v_2, v_3 \rightarrow Z'(p_1)$.

Internal momenta: $P_1 = k_3 + k_2, P_2 = k_3 + k_2 + k_1$.

$$\mathcal{A}_c^{\mu\nu\rho} = \frac{\bar{u}(p_2) \gamma^\rho (\not{P}_2 + m_\chi) \gamma^\mu (\not{P}_1 + m_\chi) \gamma^\nu u(k_3)}{D_1^{(c)} D_2^{(c)}} \quad (8)$$

$$D_1^{(c)} = (k_3 + k_2)^2 - m_\chi^2 = 2k_2 \cdot k_3 + m_{Z'}^2, \quad D_2^{(c)} = (k_3 + k_2 + k_1)^2 - m_\chi^2.$$

Diagram (d): $Z'(k_2) \rightarrow v_1, v_2 \rightarrow Z'(p_1), Z'(k_1) \rightarrow v_3$.

Internal momenta: $P_1 = k_3 + k_2, P_2 = k_3 + k_2 - p_1$.

$$\mathcal{A}_d^{\mu\nu\rho} = \frac{\bar{u}(p_2) \gamma^\mu (\not{P}_2 + m_\chi) \gamma^\rho (\not{P}_1 + m_\chi) \gamma^\nu u(k_3)}{D_1^{(d)} D_2^{(d)}} \quad (9)$$

$$D_1^{(d)} = (k_3 + k_2)^2 - m_\chi^2 = 2k_2 \cdot k_3 + m_{Z'}^2, \quad D_2^{(d)} = (k_3 + k_2 - p_1)^2 - m_\chi^2.$$

Diagram (e): $v_1 \rightarrow Z'(p_1), Z'(k_2) \rightarrow v_2, Z'(k_1) \rightarrow v_3$.

Internal momenta: $P_1 = k_3 - p_1, P_2 = k_3 - p_1 + k_2$.

$$\mathcal{A}_e^{\mu\nu\rho} = \frac{\bar{u}(p_2) \gamma^\mu (\not{P}_2 + m_\chi) \gamma^\nu (\not{P}_1 + m_\chi) \gamma^\rho u(k_3)}{D_1^{(e)} D_2^{(e)}} \quad (10)$$

$$D_1^{(e)} = (k_3 - p_1)^2 - m_\chi^2 = m_{Z'}^2 - 2 k_3 \cdot p_1, \quad D_2^{(e)} = (k_3 - p_1 + k_2)^2 - m_\chi^2.$$

Diagram (f): $v_1 \rightarrow Z'(p_1)$, $Z'(k_1) \rightarrow v_2$, $Z'(k_2) \rightarrow v_3$.

Internal momenta: $P_1 = k_3 - p_1$, $P_2 = k_3 - p_1 + k_1$.

$$\mathcal{A}_f^{\mu\nu\rho} = \frac{\bar{u}(p_2) \gamma^\nu (\not{P}_2 + m_\chi) \gamma^\mu (\not{P}_1 + m_\chi) \gamma^\rho u(k_3)}{D_1^{(f)} D_2^{(f)}} \quad (11)$$

$$D_1^{(f)} = (k_3 - p_1)^2 - m_\chi^2 = m_{Z'}^2 - 2 k_3 \cdot p_1, \quad D_2^{(f)} = (k_3 - p_1 + k_1)^2 - m_\chi^2.$$

Note the following pairings among the propagator denominators: diagrams (a) and (c) share $D_2 = (k_3 + k_1 + k_2)^2 - m_\chi^2$; diagrams (a) and (b) share $D_1 = (k_3 + k_1)^2 - m_\chi^2$; diagrams (c) and (d) share $D_1 = (k_3 + k_2)^2 - m_\chi^2$; diagrams (e) and (f) share $D_1 = (k_3 - p_1)^2 - m_\chi^2$.

4 Spin/Polarization-Averaged Squared Amplitude

The spin- and polarization-averaged squared amplitude is

$$\overline{|\mathcal{M}|^2} = \frac{1}{18} g_D^6 \sum_{\text{pol, spins}} \left| \sum_{i=a}^f \mathcal{A}_i^{\mu\nu\rho} \epsilon_\mu(k_1) \epsilon_\nu(k_2) \epsilon_\rho^*(p_1) \right|^2, \quad (12)$$

where the factor $1/18 = 1/(2 \times 3 \times 3)$ averages over the initial χ spin (2 states) and the two initial Z' polarizations (3 states each). The massive vector polarization sum is

$$\sum_\lambda \epsilon_\alpha^{(\lambda)*}(k) \epsilon_\beta^{(\lambda)}(k) = -g_{\alpha\beta} + \frac{k_\alpha k_\beta}{m_{Z'}^2}, \quad (13)$$

and the fermion spin sums are $\sum_s u^s(p) \bar{u}^s(p) = \not{p} + m$.

The squared amplitude decomposes into 36 interference terms (6 diagonal + 15 off-diagonal $\times 2$), each involving a single Dirac trace of rank up to 12 Lorentz indices. These are evaluated with **FeynCalc** [3] and expressed in terms of the five invariants $\{s_{12}, s_{13}, t_1, t_2, t_3\}$ and the masses $m_\chi, m_{Z'}$.

5 Non-Relativistic Limit and Thermal Average

In the non-relativistic (NR) limit all initial particles are at rest in the centre-of-mass frame:

$$k_1^\mu = (m_{Z'}, \mathbf{0}), \quad k_2^\mu = (m_{Z'}, \mathbf{0}), \quad k_3^\mu = (m_\chi, \mathbf{0}). \quad (14)$$

The total CM energy is $\sqrt{s} = 2m_{Z'} + m_\chi$, and the invariants reduce to

$$s_{12} = 4m_{Z'}^2, \quad s_{13} = (m_{Z'} + m_\chi)^2, \quad (15)$$

$$t_1 = t_2 = 2m_{Z'}^2 - 2m_{Z'} E_{Z'}^f, \quad t_3 = m_\chi^2 + m_{Z'}^2 - 2m_\chi E_{Z'}^f, \quad (16)$$

where the final-state Z' energy is

$$E_{Z'}^f = \frac{s + m_{Z'}^2 - m_\chi^2}{2\sqrt{s}}. \quad (17)$$

Crucially, since all initial three-momenta vanish, every scalar product is independent of the scattering angle θ . Therefore $\overline{|\mathcal{M}|^2}_{\text{NR}}$ is isotropic and the angular integral is trivial:

$$\int_{-1}^1 d\cos\theta \overline{|\mathcal{M}|^2}_{\text{NR}} = 2 \overline{|\mathcal{M}|^2}_{\text{NR}}. \quad (18)$$

For an v -independent cross section, the NR threshold value equals the thermally averaged quantity:

$$\langle \sigma v^2 \rangle_{\text{NR}} = (\sigma v^2)|_{\text{threshold}}. \quad (19)$$

6 Result: $\langle \sigma v^2 \rangle$ and Δ_3

Using the $3 \rightarrow 2$ body phase space from Eq. (A11) of Ref. [2], the thermally averaged cross section in the NR limit is

$$\langle \sigma v^2 \rangle = g_D^6 \frac{|\mathcal{M}|_{\text{NR}}^2 \sqrt{\lambda(s, m_{Z'}^2, m_\chi^2)}}{64\pi m_{Z'}^2 m_\chi s} \quad (20)$$

where $s = (2m_{Z'} + m_\chi)^2$, $\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2ac - 2bc$ is the Källén function, and $|\mathcal{M}|_{\text{NR}}^2$ is the spin/polarization-averaged squared amplitude with g_D^6 factored out, evaluated at threshold. The factor of 2 from the angular integral has been combined with the $1/(128\pi)$ in the phase space formula to yield $1/(64\pi)$.

The fudge factor Δ_3 of Ref. [1] is defined via

$$\langle \sigma v^2 \rangle = \frac{\Delta_3 \alpha_X^3}{m_\chi^5}, \quad (21)$$

and is extracted from Eq. (20) as

$$\Delta_3(r) = \langle \sigma v^2 \rangle \times \frac{m_\chi^5}{\alpha_X^3}, \quad r \equiv \frac{m_\chi}{m_{Z'}}. \quad (22)$$

The explicit result from the FeynCalc computation is

$$\Delta_3(r) = \frac{4\pi^2 r^5 \sqrt{4r(r+2)/3+1} \mathcal{P}(r)}{(r+2)^3 (2r+1)^4 [2r^2(r+2) + r - 1]^2} \quad (23)$$

where the numerator polynomial is

$$\mathcal{P}(r) = 2r[r(r(2r(r(8r(10r+71) + 1683) + 2798) + 6107) + 4722) + 2335) + 578] + 195. \quad (24)$$

Selected numerical values:

$r = m_\chi/m_{Z'}$	1	2	10	100
$\Delta_3(r)$	54.04	144.02	1610.5	21931.3

Short Discussion. I find that $\Delta_3 \gg 1$ at large r which is seemingly in tension with the “order one” characterization of Ref. [1]. However, I think this relationship makes physical sense. In [1], Δ_3 is simply estimated it by dimensional analysis using a single scale m_χ . However, the process involves two scales (m_χ and $m_{Z'}$), and the propagator denominators scale as $\sim m_{Z'} m_\chi$ rather than m_χ^2 , such that when $m_{Z'} \ll m_\chi$, we have an enhancement in powers of r .

References

- [1] A. Berlin, D. Hooper, and G. Krnjaic, “PeV-Scale Dark Matter as a Thermal Relic of a Decoupled Sector,” *Phys. Lett. B* **760**, 106 (2016), arXiv:1602.08490.
- [2] A. Berlin, D. Hooper, and G. Krnjaic, “Thermal Dark Matter From A Highly Decoupled Sector,” *Phys. Rev. D* **94**, 095019 (2016), arXiv:1609.02555.
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